

Device: cuda

Torch dtype: torch.float32

M: 48

Nodes: 117,649

Undirected edges: 345,744

/tmp/ipykernel_806/1672803553.py:130: UserWarning: Sparse invariant checks are implicitly disabled. Memory errors (e.g. SEGFault) will occur when operating on a sparse tensor which violates the invariants, but checks incur performance overhead. To silence this warning, explicitly opt in or out. See ``torch.sparse.check_sparse_tensor_invariants.__doc__`` for guidance. (Triggered internally at /pytorch/aten/src/ATen/Context.cpp:760.)

```
L = torch.sparse_coo_tensor(indices, values, (N, N), device=device).coalesce()
```

Source coordinate: (np.int64(24), np.int64(24), np.int64(24))

Source index: 58824

CG iter 1 | relative residual 4.082e-01

CG iter 100 | relative residual 1.888e-05

CG converged at iter 119 | relative residual 9.317e-07

CG elapsed: 0.06 s

--- Green kernel fit ---

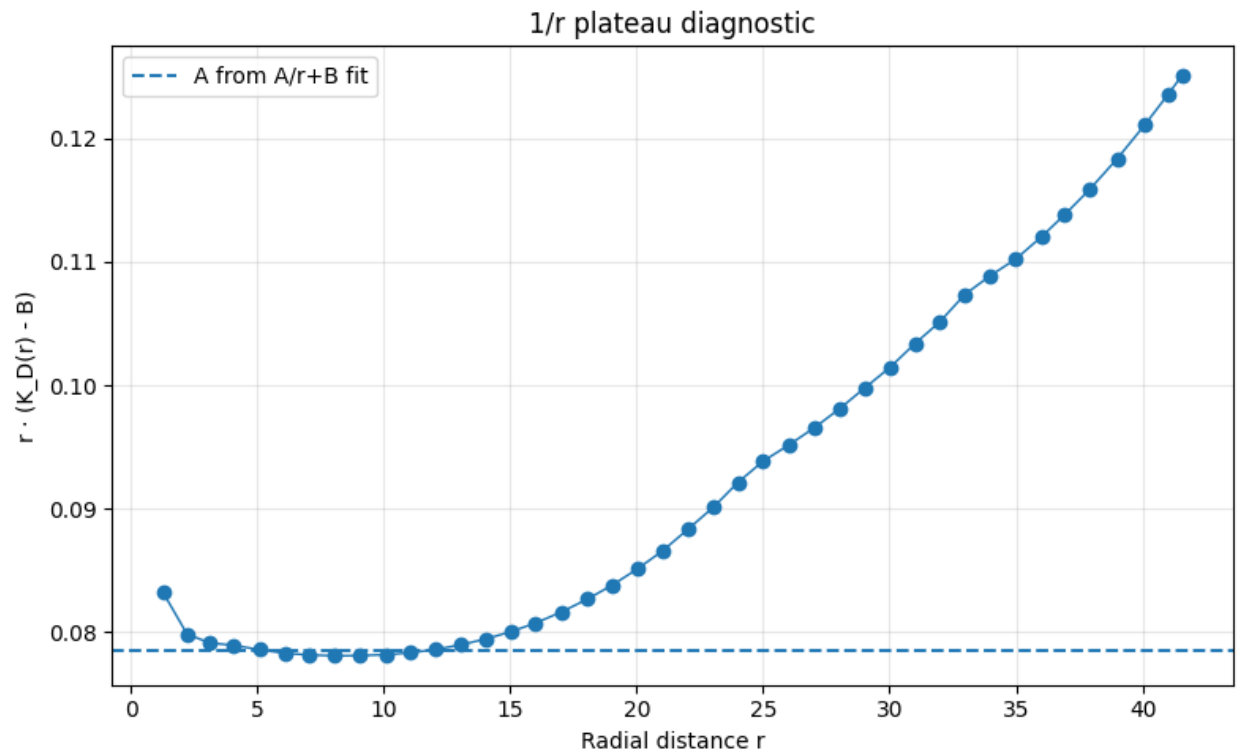
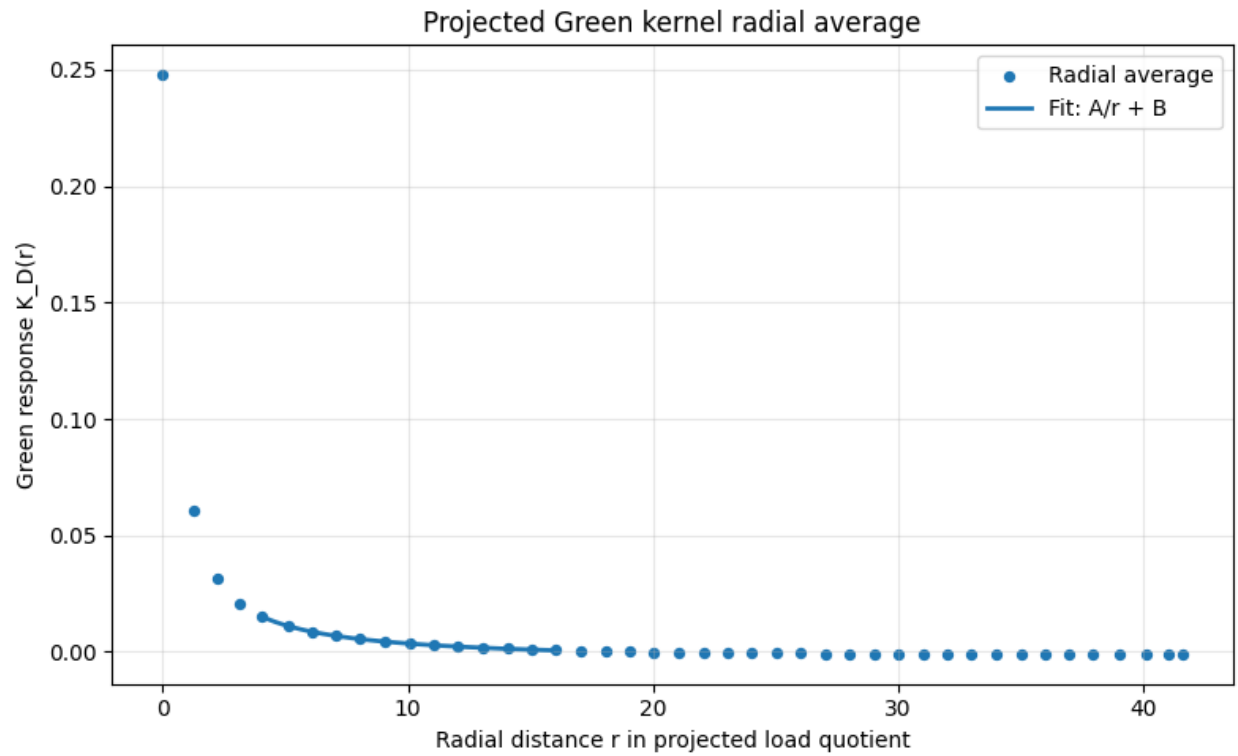
$$K(r) \approx A/r + B$$

A = 7.85345551e-02

B = -4.31376118e-03

R² = 0.99982638

Fit window: r in [4, 15.84]



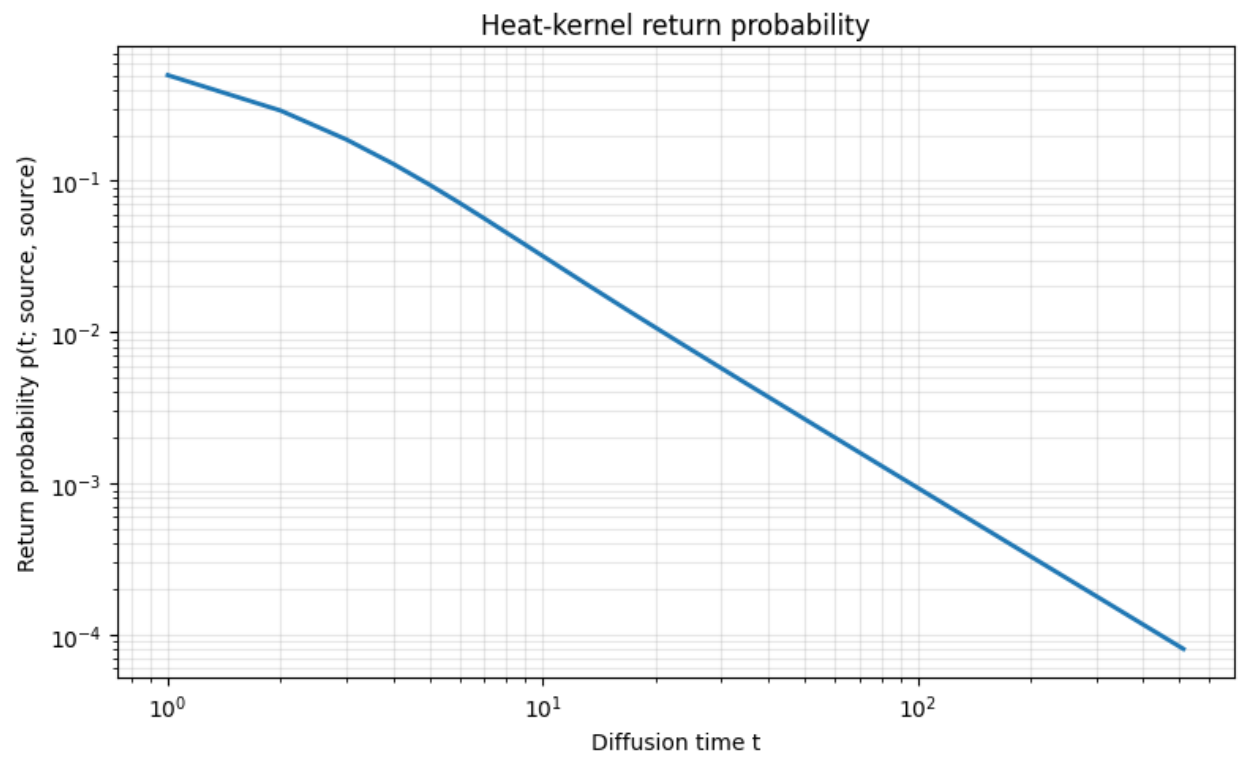
Heat diffusion elapsed: 0.03 s

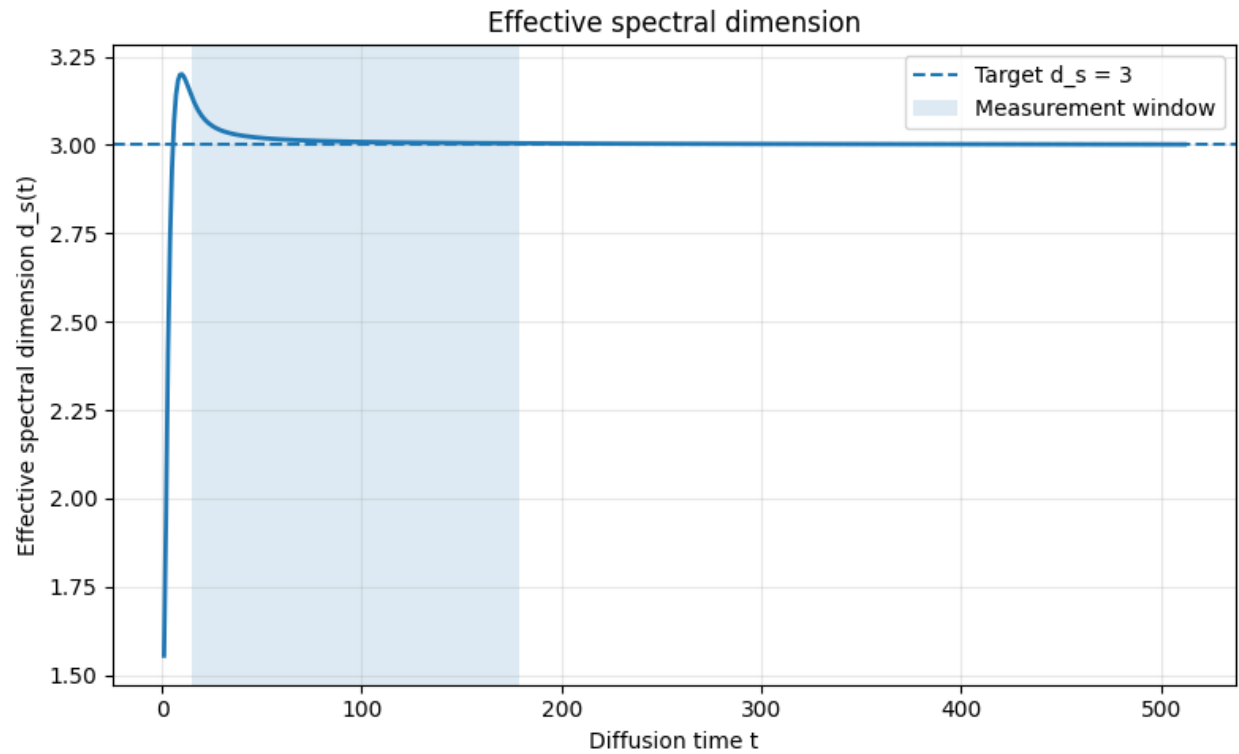
--- Heat-kernel spectral dimension ---

Window: t in $[15, 179]$

Median $d_s \approx 3.0088$

Mean $d_s \approx 3.0171$





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SUMMARY

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Green $1/r$ fit R^2 : 0.99982638 -> PASS

Median spectral dimension: 3.0088 -> PASS

Output directory:

/content/pregeo_green_kernel_results

Files written:

- cg_convergence.csv
- green_kernel_fit_summary.csv
- green_kernel_plateau_diagnostic.png

- green_kernel_radial_average.csv
- green_kernel_radial_fit.png
- heat_kernel_spectral_dimension.csv
- heat_return_probability.png
- spectral_dimension_estimate.png

Manuscript-safe interpretation:

This numerical experiment computes the Green response $K_D = L_{\text{grav}}^+$ on a projected three-dimensional admissible load quotient. The radial Green response is tested against $K_D(r) = A/r + B$, while the heat-kernel return probability is used to estimate the effective spectral dimension. A simultaneous $1/r$ Green window and $d_s \approx 3$ heat window supports, but does not prove, the claim that the projected load sector has a Newtonian weak-field continuum limit. A formal proof must still establish that the admissible quotient selected by the Hodge-Dirac/least-action construction has three-dimensional volume growth and heat-kernel bounds in the large-system limit.